https://xtremepape.rs/

Question	Answer	Marks
1	**Defining the problem**	
	A is the independent variable and *x* is the dependent variable **or** vary *A* and measure *x*	**1**
	keep *m* constant	**1**
	Methods of data collection	
	labelled diagram of workable experiment including: • spring fixed at one end (to a support) • load attached to the other end • labelled load/mass	**1**
	clamped labelled ruler perpendicular by eye **and** ruler positioned close to the spring	**1**
	method to determine *x*: record initial position, record final position **and** subtraction **or** record length of stretched spring, original length of spring **and** subtraction	**1**
	use of a micrometer/calipers to determine diameter *d* of wire	**1**
	Method of Analysis	
	plots a graph of lg *x* against lg *A* or equivalent	**1**
	n = gradient **and** relationship valid if a straight line	**1**
	$\gamma = \dfrac{mgw^3N}{\rho \times 10^{y\text{-intercept}}}$	**1**

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1	**Additional detail including safety considerations**	**6**
	D1 use safety goggles/safety screen <u>to prevent injury to eyes from (moving) spring/load</u> **or** use cushion/sand box <u>in case load falls</u>	
	D2 keep constant <u>all</u> of: • w • N • ρ (or use same material for wire)	
	D3 measure mass <u>of load/m</u> with a balance **or** use newton meter to measure weight of load	
	D4 use ruler/calipers to measure the width of the <u>spring/w</u>	
	D5 use the equation ρ = mass of metal / volume to calculate ρ	
	D6 method to measure w, e.g. w measured to the outside and subtract d **or** method to measure ρ, e.g. mass (balance) and volume (measuring cylinder)	
	D7 repeat measurement of x (for the same A) and average	
	D8 repeat measurements of either d or w made in different places/directions and average	
	D9 use of $A = \pi d^2 / 4$ to determine A of <u>wire</u>	
	D10 use of set square to check that clamped ruler is vertical to bench **or** use of set square/pin as a fiducial mark to read positions (on the ruler)	

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